

Non-Calculator Questions

Right-Angled Triangles

Pythagoras Theorem / SOHCAHTOA / Angles of Elevation & Depression / Exact Trig Values

Medium (4 questions)	/15
Hard (3 questions)	/11
Very Hard (4 questions)	/18
Total Marks	/44

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Medium Questions

1 (a) x is an acute angle and $\cos x = \frac{\sqrt{3}}{2}$.

Find the value of $\sin x$

(2 marks)

(b) Find the value of $\cos(180^\circ + x)$

(1 mark)

2 Here is a rectangle.

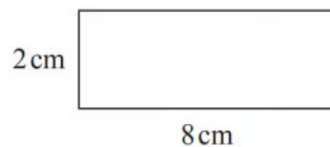


Diagram **NOT**
accurately drawn

The 8-sided shape below is made from 4 of these rectangles and 4 congruent right-angled

triangles.

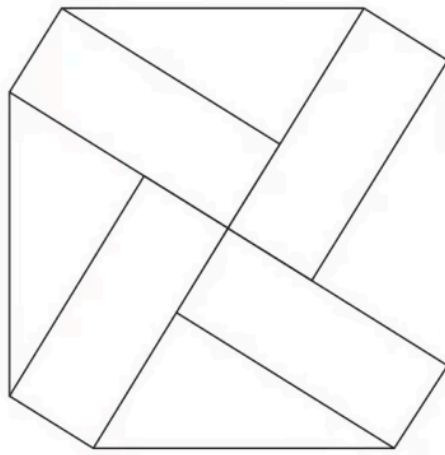


Diagram **NOT**
accurately drawn

Work out the perimeter of the 8-sided shape.

You must show all your working.

(5 marks)

3 Triangle ABC has perimeter 20 cm.

$$AB = 7 \text{ cm.}$$

$$BC = 4 \text{ cm.}$$

By calculation, deduce whether triangle ABC is a right-angled triangle.

(4 marks)

4

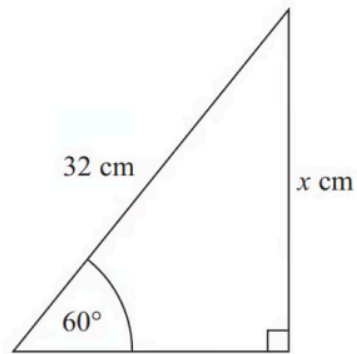


Diagram **NOT**
accurately drawn

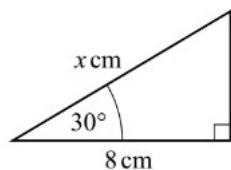
Calculate the value of x .

Give your answer as an exact value.

(3 marks)

Hard Questions

1



NOT TO
SCALE

Find the exact value of x .

$x = \dots\dots\dots$

(4 marks)

2 The length of one side of a rectangle is 12 cm .

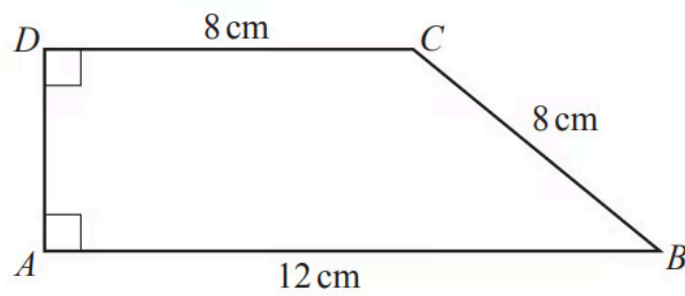
The length of the diagonal of the rectangle is 13 cm .

Calculate the area of the rectangle.

$\dots\dots\dots\text{ cm}^2$

(3 marks)

3 Calculate the area of this trapezium.



NOT TO
SCALE

Give your answer as an exact value.

..... cm^2

(4 marks)

Very Hard Questions

1

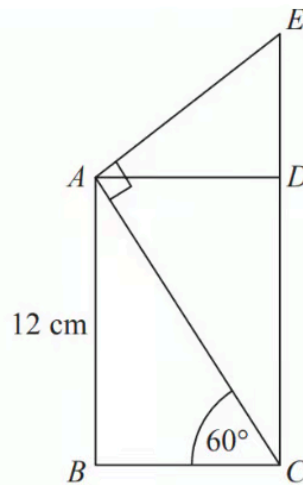


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CDE is a straight line.

$AB = 12$ cm Angle

$ACB = 60^\circ$

Angle $EAC = 90^\circ$

Calculate the length of CE .

You must show all your working.

(4 marks)

2 The table shows some values of x and y that satisfy the equation $y = a \cos x^\circ + b$

x	0	30	60	90	120	150	180
y	3	$1 + \sqrt{3}$	2	1	0	$1 - \sqrt{3}$	-1

Find the value of y when $x = 45$

(4 marks)

3 Simplify $\frac{2 \sin 45^\circ - \tan 45^\circ}{4 \tan 60^\circ}$

Give your answer in the form $\frac{\sqrt{a} - \sqrt{b}}{c}$ where a , b and c are integers.

(4 marks)

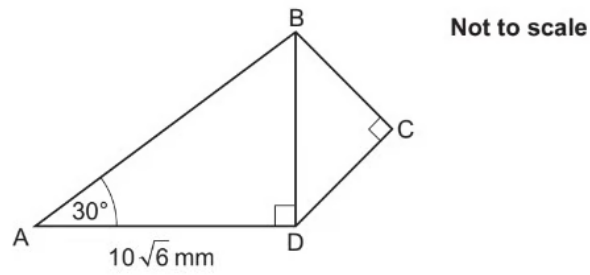
4 ADB and BCD are right-angled triangles.

$$BC = CD.$$

$$AD = 10\sqrt{6} \text{ mm}$$

$$\text{Angle BAD} = 30^\circ.$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$



Work out the length of BC.

.....mm

(6 marks)